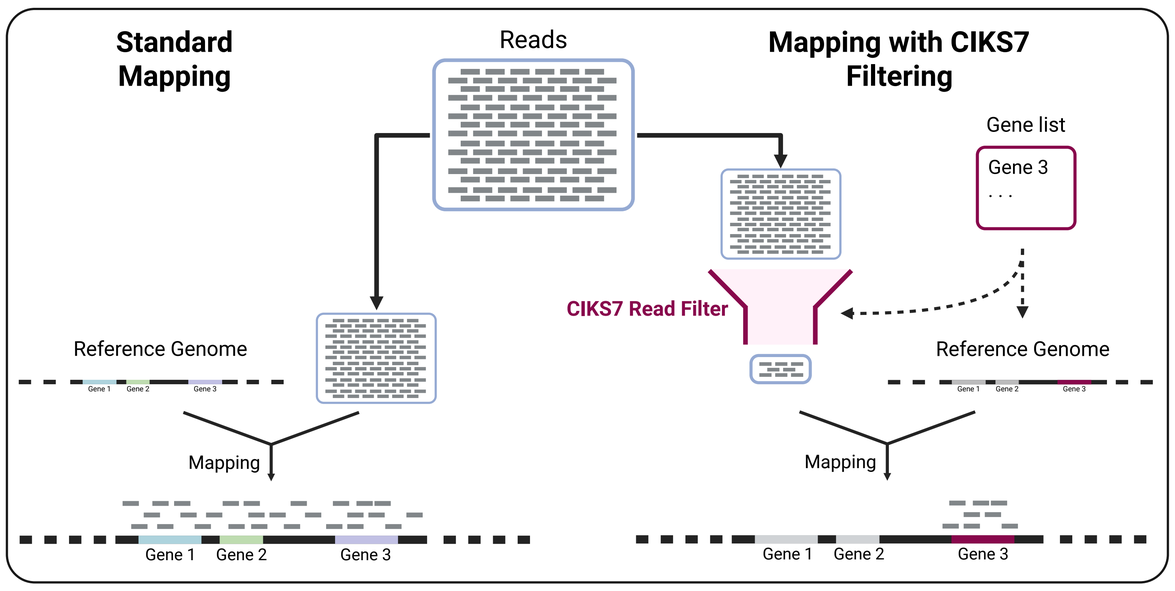
CIKS7 - Counting and Identification using K-mer based Selection

Aisling Moloney, Benedikt Leitz

MOTIVATION

- Standard read mapping aligns **all reads** → computationally expensive
- CIKS7 uses **k-mer matching** to pre-filter reads
- Targets **genes of interest before alignment**
- Strongly reduces reads passed to mapping
- Lowers **computational cost of downstream analysis**



METHODS

Method Overview

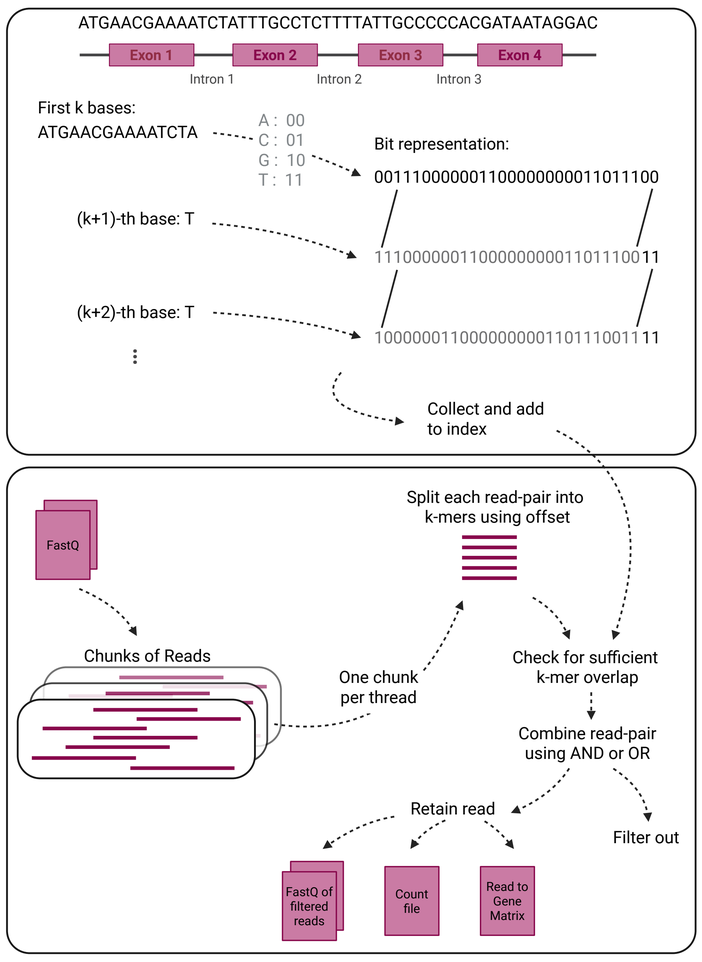
- k-mer–based indexing of target genes for fast filtering
- Processes **paired-end FASTQ reads**

Key Parameters

- k-mer length
- offset
- threshold

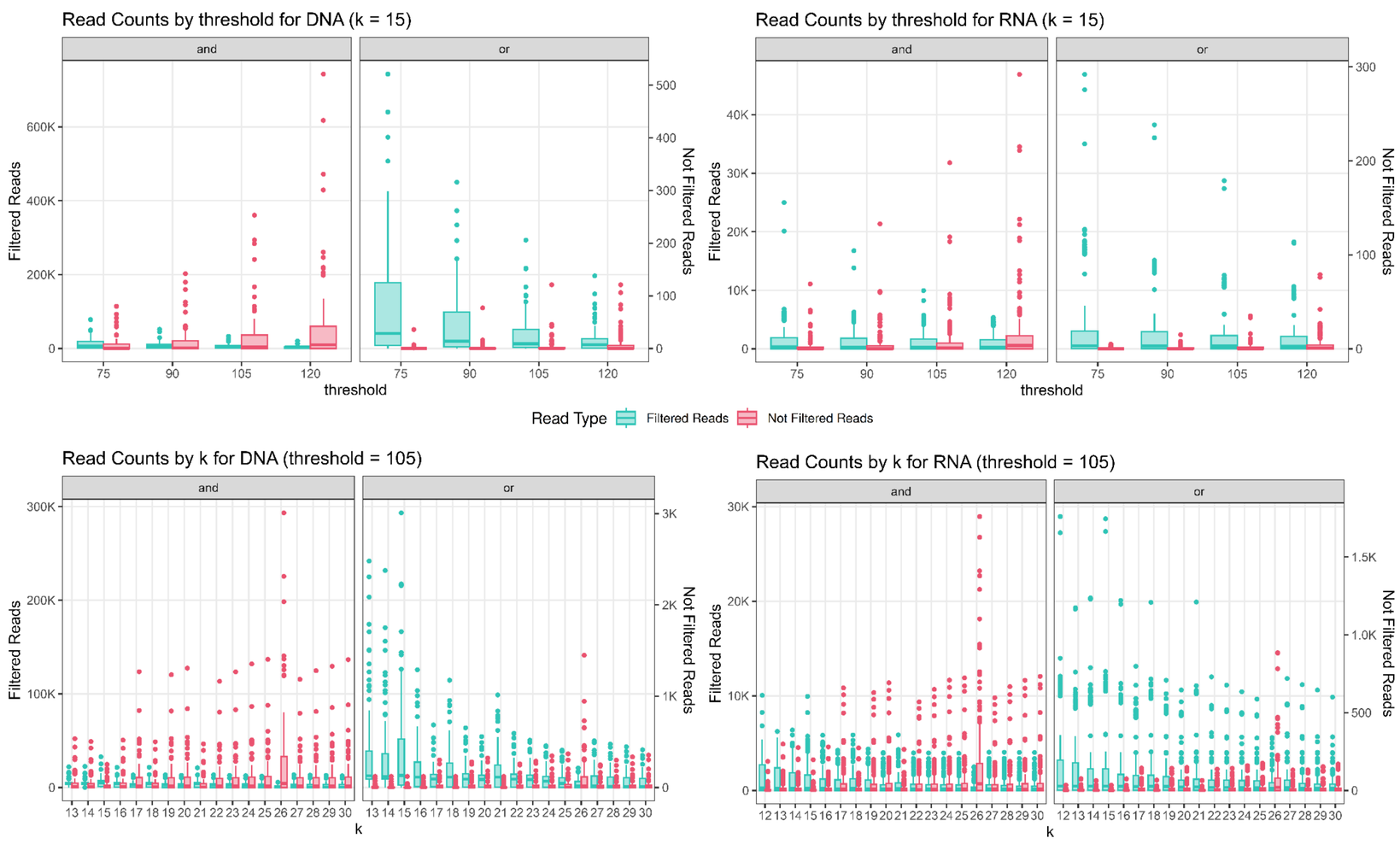
Run Modes

- **DNA vs. RNA:** gene vs. transcript-based index
- **AND vs. OR:** both reads vs. one read must match

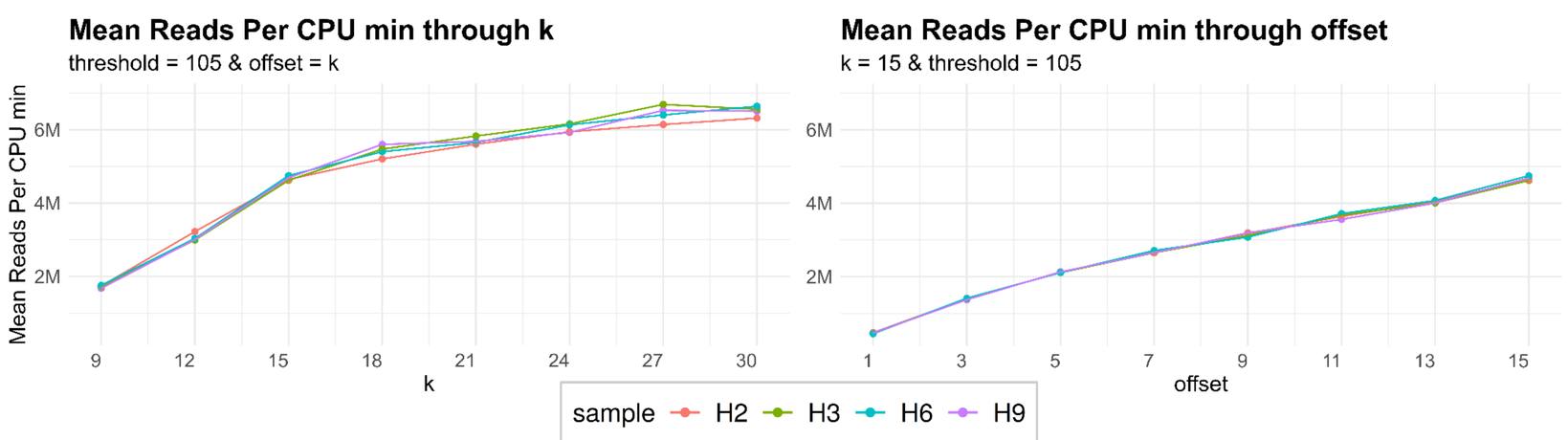


CIKS7 workflow: index creation (top), read processing (bottom)

RESULTS



Boxplots showing the number of filtered (retained) and not filtered (discarded but mapped by Minimap2) reads across different thresholds for k = 15 and offset = 15 and across different values of k with threshold = 105 and offset = 15 for both DNA and RNA modes. Note that the scales differ.
Each dot represents the read count for a gene. ENSSSCG00000018081 and ENSSSCG00000035362 were excluded.



Mean processed reads per CPU minute both across different values of k and offset for the four testes samples H2, H3, H6, H9 (all from pig aorta tissue).

Computational Performance

- Processes ~**13.5 million reads per minute**
- Runtime scales **linearly with input size**
- **k and offset** strongly affect performance (smaller values → more k-mers → slower)
- Memory ~5 GB for typical settings (higher for very small offsets)
- Best balance: k ≥ 15 and offset ≥ 5

RESULTS

Filter Quality

- Removes **>95% of reads** while retaining nearly all reads from target genes
- Maintains **high mapping agreement** with reference mappings
- **OR mode:** retains more reads, minimizes missed reads
- **AND mode:** stricter filtering, but may miss some mapped reads
- **RNA mode:** strongly reduces retained reads by focusing on exons, with similar mapping accuracy

Recommended Parameters

- Parameter choice strongly affects filtering performance
- **k = 15** provides a good balance of speed and accuracy
- **Offset = k** avoids unnecessary runtime overhead
- **Threshold:** use **medium values (~50–66% of read length)**
- Balances retained reads and risk of missing mapped reads

CONCLUSION

- Fast **k-mer based read filter** for targeted regions
- Processes ~**13.5 million reads per minute**
- Removes **>95% of reads** while retaining nearly all relevant reads
- Parameter choices balance **sensitivity and performance**

Created with BioRender Poster Builder